Franklin Open 17 (2026) 100741

Contents lists available at ScienceDirect

Franklin Open

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A feature-fusion ensemble deep learning framework for accurate and automated leukaemia diagnosis

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ARTICLE INFO

Keywords:
Leukaemia detection
Ensemble learning
Convolutional neural network (CNN)
Modified CNN
ResNet50
Feature fusion
HOG
LDP

ABSTRACT

Leukaemia is a haematological malignancy that requires early and accurate diagnosis for effective treatment. Conventional manual techniques for examining smear samples are time-consuming and prone to human error. This study introduces an ensemble architecture for leukaemia diagnosis. The proposed design, a modified Convolutional Neural Network (MCNN), integrates hierarchical deep features from ResNet50 with manually constructed descriptors, specifically Local Directional Patterns (LDP) and Histogram of Oriented Gradients (HOG), to enhance feature extraction and overcome inherent CNN limitations such as overfitting. The combination of baseline CNN, ResNet50, and MCNN yielded the best diagnostic performance, achieving 96.50% accuracy, 96.30% precision, 96.70% recall, and a 96.50% F1-score over 50 training epochs, achieving better performance than individual MCNN (85.47% accuracy) and ResNet50 (88.50% accuracy) baselines. The ensemble of deep hierarchical features and handcrafted descriptors within the CNN framework simplifies the differentiation between leukaemic cells across various datasets. This ensemble framework is a generalised model that can significantly support clinical decision support systems, particularly in resource-limited healthcare environments.

1. Introduction

Acute lymphoblastic leukaemia is the most prevalent childhood cancer and the leading cause of mortality in many parts of the world. Despite the existence of the medical treatment process, 10%–15% of patients do not achieve remission. Effective chemotherapy is hindered by drug resistance and diverse patient responses [1,2]. Hence, precise detection is crucial for proper treatment and reducing mortality risk [3,4]. Conventional diagnosis relies on the manual examination of blood smear samples, which is time-consuming and prone to human error by expert pathologists, potentially leading to devastating effects. To overcome these limitations, automated systems based on deep learning architectures have been developed [5]. CNN have proven their architectural success in classifying blood disorders from microscopic images [6,7]. Recently, hybrid models that combine different models with traditional feature extraction have significantly improved detection accuracy by addressing the specific limitations of traditional CNN baseline models, such as overfitting on small datasets [8,9]. This study integrates a unified framework for leukaemia diagnosis using a multimodal feature Fusion Ensemble [10]. By integrating ResNet50 with hand-crafted descriptors (HOG and LDP) and a baseline CNN, the proposed model enhances feature discriminability and generalisation of robust clinical decision support.

2. Literature survey

Recent developments in medical image analysis have shifted towards deep learning model architectures to improve ALL detection. Table 1 summarises the key methodologies and contributions in this domain, highlighting the progression from standard CNNs to hybrid detection pipelines to the proposed method.

3. Proposed methodology

In this study, we propose a robust multimodal feature fusion ensemble framework for automated leukaemia detection. The framework extracts deep semantic features from ResNet50 using handcrafted descriptors (HOG, LDP). The proposed architecture in Fig. 1 combines three distinct classifiers, a baseline CNN, a standalone ResNet50, and a modified CNN (Hybrid Fusion), within a single weighted ensemble framework to balance a high-level abstraction with low-level texture details.

3.1. Preprocessing

To ensure data consistency and robustness, raw blood smear images were subjected to pre-processing steps, including uniform resizing,

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https://doi.org/10.1016/j.fraope.2026.100741
Received 3 December 2025; Received in revised form 23 March 2026; Accepted 17 August 2026
Available online 22 August 2026
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Table 1
Literature survey

Ref	Methodology	Key benefits	Short description
[7]	CNN for leukaemia cell identification	Automated identification of leukaemia; reduces reliance on human expertise	Demonstrated the success of CNNs in identifying leukaemia cells from microscopic images, supporting diagnostic automation.
[11]	CNN refinement for leukaemia subtype classification	Improved classification performance	Extended CNN to classify various leukaemia subtypes with impressive results.
[12,13]	Integration of YOLO with CNN and texture analysis	High-speed processing; useful for time-sensitive diagnosis	Combined YOLO real-time detection with CNN and traditional texture features to improve the classification accuracy.
[8]	MI-based hybrid model integrating various feature extraction methods	Reduces overfitting; less dependent on large labelled datasets	Developed a model that integrates classical features with CNN, improving performance and robustness.
[14]	YOLO for object detection	Real-time diagnosis; high efficiency	YOLO processes entire images in one pass, enabling rapid diagnostic decisions.
[15]	Optimisation of CNN using genetic algorithms and particle swarm optimisation	Enhances model performance and efficiency	Applied metaheuristic algorithms to optimise CNN architectures in blood disorder classification.
[16]	Multi-view and self-supervised learning for image analysis	Greater generalisation; reduced need for large labelled datasets	Applied advanced learning techniques to boost model robustness and diagnostic capacity in medical image analysis.
[17]	YOLO-based detection in recent studies	High diagnostic relevance; continues to evolve	Highlights the ongoing contributions of YOLO in enhancing detection systems in medical image processing.

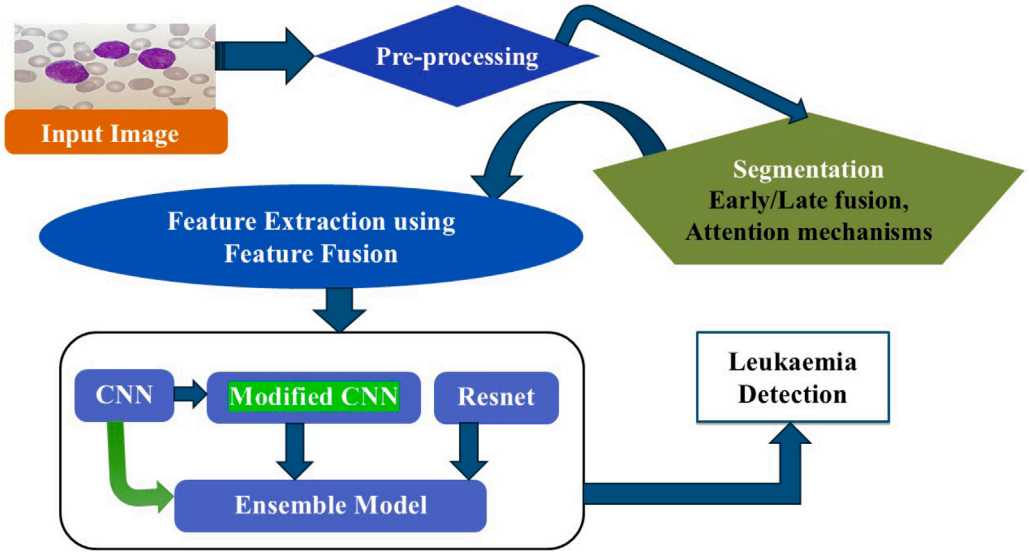


Fig. 1. Proposed block diagram of leukaemia identification using multi-model feature fusion ensemble learning.

histogram equalisation for contrast normalisation, and geometric data augmentation (rotation, flipping, and zooming) to overcome overfitting.

3.1.1. Ensemble model framework

The proposed architecture is based on an ensemble learning strategy that integrates three complementary deep learning models: a Baseline Convolutional Neural Network (CNN), ResNet50, and a Modified CNN. The ensemble operates using a weighted soft-voting mechanism. The optimised weighted coefficients of each model contributed to the architecture.

Let $P_i(c \mid x)$ denote the posterior probability of class c for an input image x predicted by the ith model. The prediction score of the proposed model S_c is calculated as follows:

$$S_c = \alpha P_1(c \mid x) + \beta P_2(c \mid x) + \gamma P_3(c \mid x) \tag{1}$$

where α, β, and γ are empirically determined weight coefficients obtained via a grid search on the validation dataset, subject to the constraint:

$$\alpha + \beta + \gamma = 1 \tag{2}$$

The grid search was conducted by iterating through potential weight combinations from 0.1 to 0.8 with a step size of 0.1 for each coefficient. The performance of an ensemble was measured with the help of the validation accuracy, considering every legal combination that satisfies

the sum constraint. The combination with the highest metric was then chosen as the best weighting setup of the final model prediction.

The final predicted class label y is determined as follows:

$$y = \arg\max_{c}(S_c) \tag{3}$$

This probabilistic fusion strategy reduces the prediction variance and increases the misclassification caused by the individual biases of the component models, resulting in optimum performance.

3.1.2. Architectural configurations

In order to have reproducibility, the exact configurations of the base networks are explained below:

Baseline CNN: This model is a baseline that has 8 sequential convolutional blocks. All the blocks use a 2D Convolutional layer with 3×3 kernel and stride 1 with Batch normalisation to stabilise the learning process. Non-linearity is added by the use of the ReLU (Rectified Linear Unit) activation function. Max Pooling layers with a window of 2×2 reduce dimensions in the spatial dimensions. The rate of Dropout regularisation 0.3 is used after the pooling layers to reduce the issue of overfitting on the training data. The final layer of the network is a dense layer that has been completely connected and culminates into a Softmax classifier.

Modified CNN: The Modified CNN is based on the baseline structure, but specific to have an extended feature vector. It implements the combination of deep semantic infusion of the terminal average pooling layer of ResNet50, as well as the handcrafted HOG and LDP texture arrays. Two dense layers of 512 and 256 neuron each are fed on this fused 1D feature vector. Both thick layers employ ReLU activations with a severe Dropout rate of 0.5 to manage the dimensionality of the concatenated feature space which is high with the last Softmax classification layer.

Algorithm 1 Hybrid Multi-modal Ensemble for Leukaemia Classification

Require: Raw microscopic blood smear image I
Ensure: Predicted class label $y \in \{\text{ALL}, \text{HEM}\}$

procedure LEUKEMIA_CLASSIFICATION(I)
  $I_{\text{pre}} \leftarrow \text{Preprocess}(I)$ ▷ Resize, Normalise, Augment
  ▷ **Path A: Baseline CNN**
  $P_1 \leftarrow \text{Softmax}(\text{CNN}(I_{\text{pre}}))$
  ▷ **Path B: Deep Semantic Feature Extraction**
  $f_{\text{Res}} \leftarrow \text{ResNet50_ext}(I_{\text{pre}})$
  $P_2 \leftarrow \text{Softmax}(\text{Classifier}(f_{\text{Res}}))$
  ▷ **Path C: Hybrid Feature Fusion (Modified CNN)**
  $v_{\text{HOG}} \leftarrow \text{ExtractHOG}(I_{\text{pre}})$
  $v_{\text{LDP}} \leftarrow \text{ExtractLDP}(I_{\text{pre}})$
  $v_{\text{Res}} \leftarrow \text{ResNet50_ext}(I_{\text{pre}})$
  $v_{\text{concat}} \leftarrow \text{Concatenate}(v_{\text{HOG}}, v_{\text{LDP}}, v_{\text{Res}})$
  $v_{\text{fused}} \leftarrow \text{DenseBlock}(v_{\text{concat}})$ ▷ Dimensionality Reduction
  $P_3 \leftarrow \text{Softmax}(\text{Classifier}(v_{\text{fused}}))$
  ▷ **Ensemble Fusion**
  $S_c \leftarrow \alpha P_1 + \beta P_2 + \gamma P_3$
  $\hat{y} \leftarrow \arg\max_c(S_c)$
  return $\hat{y}$
end procedure

4. Result and discussion

4.1. Experimental setup

Python (TensorFlow, PyTorch, and OpenCV) was used to implement these models. The experimental configuration comprised Windows 10, an Intel i7 11th Gen processor, 16 GB RAM, an NVIDIA RTX 3060 (6 GB) GPU, and a Jupyter Notebook software environment.

Table 2
Performance comparison of different models.

Metric	CNN	ResNet18	YOLOv8 [13]	Modified CNN	Proposed multimodel ensemble framework
Accuracy	77.51%	0.885	83%	85.47%	**95.3%**
Precision	75.47%	77.7%	95% (ALL) 69% (HEM)	83.81%	**96.3% (ALL)** **93.1% (HEM)**
Recall	84.03%	91.5%	81% (ALL) 77% (HEM)	97.08%	**96.6% (ALL)** **92.7% (HEM)**
F1 Score	79.49%	0.841	87% (ALL) 73% (HEM)	89.82%	**96.5% (ALL)** **92.9% (HEM)**

Dataset Partitioning: The model was evaluated on ALL-IDB and C-NMC_2019 public datasets [2,18]. In order to have a healthy assessment, the datasets were divided into training, validation and testing sets in a strict 70-15-15 split ratio. In this partitioning, we used stratified sampling so that the original class distribution in all of the three subsets would remain.

Addressing Class Imbalance: Since there is always an inherent class imbalance in medical datasets, where ALL (leukemic) samples often outnumber HEM (normal) samples, we take this into account. In order to avoid the models forming a majority-class bias, we used class-weighted categorical cross-entropy loss in the training. This approach struck the right balance to the effective training distribution, as the ensemble received the same amount of discriminative features of each of the classes which was manifested in the high recall (92.7%) and F1-score (92.9%) of the HEM class.

4.2. Model performance analysis

Each model was evaluated using its True Positive Rate (TPR), True Negative Rate (TNR), Accuracy, and F1 Score. The confusion matrices and classification reports for each model are listed in Table 2.

4.2.1. Ensemble model training graph

The training and validation accuracies are presented in Fig. 2 as a steadily rising trend, reaching a convergence of approximately 96.8% by epoch 20, as shown in the left panel. This suggests excellent generalisation to unknown data and efficient learning capabilities. The matching training and validation loss curves are shown in the right panel. They showed a consistent decrease, with the validation loss closely following the training loss during the training phase. The modest divergence of the training and validation curves indicated that the model was not overfitting and continued to perform consistently over epochs. These outcomes demonstrate the efficacy of the ensemble learning strategy in achieving high prediction accuracy while maintaining generalisation.

4.3. Ablation table for multimodel ensemble approach

Table 3 shows the ablation investigations of different model configurations that produced the recommended ensemble. The simple CNN model performed well, with an accuracy of 77.00%. The hybrid elements (HOG, LBP, and colour histogram) boosted the sensitivity to 97.08% recall. ResNet increases feature representation, thereby increasing the F1 score and accuracy. Combining detection models with 95.30% accuracy and 96.70% recall with feature extractors shows advantages in this approach. The CNN runs with hybrid features, ResNet.

The addition of hand-made descriptors is specifically aimed at the morphological aberrations of the leukemic cells that may be missed by deep models. Histogram of Oriented Gradients (HOG) is used to obtain the world wide edge directions and cell shape outlines of the

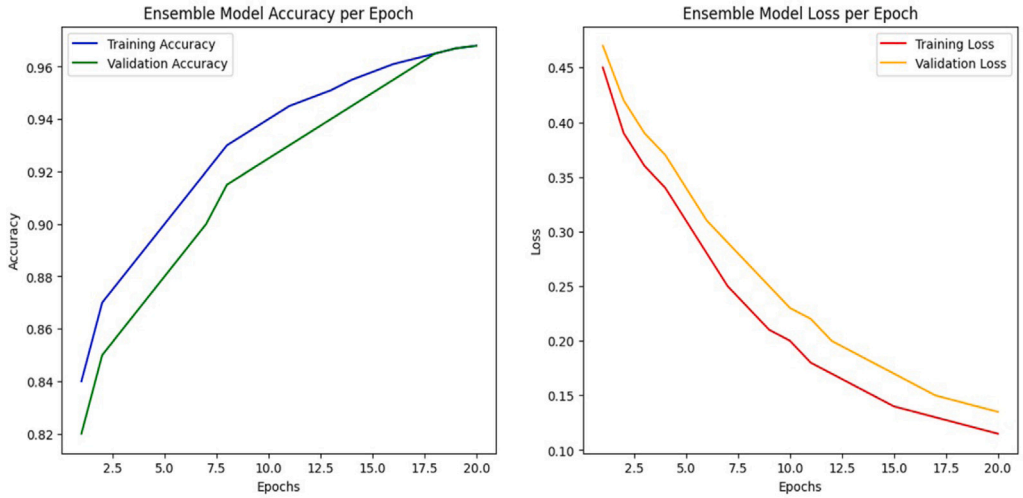


Fig. 2. Training graphs of the multimodel ensemble model.

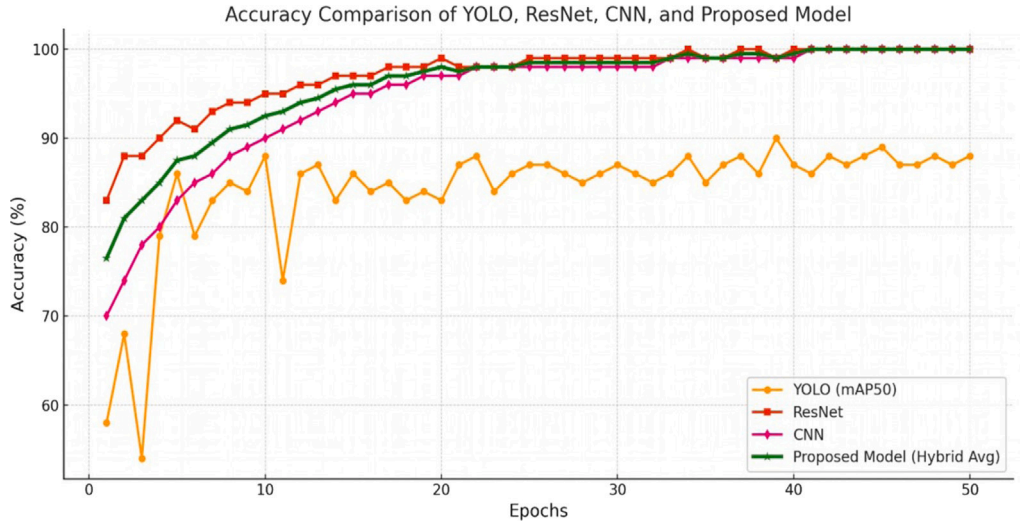


Fig. 3. Accuracy comparison.

Table 3
Ablation study table.

Configuration	CNN	HOG	LBP	ResNet50	Accuracy (%)
Baseline CNN	Yes	No	No	No	77.00
CNN + HOG + LBP	Yes	Yes	Yes	No	82.40
CNN + ResNet50	Yes	No	No	Yes	84.70
CNN + HOG + LDP + ResNet50	Yes	Yes	Yes	Yes	85.47
Full Ensemble (Proposed)	Yes	Yes	Yes	Yes	**95.30**

Table 4
Performance comparison of different models.

Model	Accuracy	Precision	Recall	F1 score
Baseline CNN	77.00%	75.00%	84.00%	79.50%
ResNet	88.50%	77.70%	91.15%	91.00%
Modified CNN	85.47%	83.81%	97.08%	89.82%
Proposed Multimodel Ensemble	**95.30%**	**96.30%**	**96.70%**	**96.50%**

cell nucleus and cytoplasm. On the contrary, Local Directional Patterns (LDP) are very sensitive to micro-textures difference and intensity changes within the cell parts. The combination of HOG (structural shape) and LDP (fine-grained texture) allows the model to have a comprehensive representation of the cell. As indicated in the ablation study, this combination of elements served to enhance the sensitivity of the baseline CNN to a recall of 97.08% which was caused by handcrafted spatial features being critical in supplying complementary information to deep semantic features being extracted by ResNet50.

4.3.1. *Analysis of varying training data*

Fig. 3 shows the training accuracy of the ResNet, modified CNN, and ensemble models for 50 epochs. The baseline CNN achieved only 77.84% accuracy owing to its limited capacity to learn only low-level features. ResNet with deep residual connections converged quickly and achieved an accuracy of 99.96%. The modified CNN, which combined deep and handcrafted features, converged faster and more accurately. The ensemble model achieved the highest accuracy of 99.99%. These results demonstrate the integration of deep learning with handcrafted

features. Its description enhances both the efficiency and reliability of leukaemia diagnosis using microscopic images (see Table 4).

4.3.2. Comparative discussion

Five models were compared: Baseline CNN, ResNet, Modified CNN, and the Proposed Ensemble. The Baseline CNN shows limited performance with 77% accuracy and a 79.5% F1 score, reflecting its weaker feature learning capacity. ResNet significantly improves performance with 88.5% accuracy and a 91% F1 score, though its lower precision indicates more false positives. The Modified CNN offers a better balance, with a strong recall of 97.08 percent and an F1 score of 89.82 percent owing to architectural improvements. The Proposed Ensemble performed best across all metrics, achieving 95.3% accuracy, 96.3% precision, 96.7% recall, and 96.5% F1 score. Statistical tests confirmed that the improvements were significant and consistent, demonstrating the strength of ensemble learning for reliable and robust classification. In addition to analysing our internal baselines, it is essential to position the proposed ensemble framework against recent external approaches. Table 1 summarises results from standard CNN-based leukaemia detection methods, including Sahlol et al. [7], which often show limited generalisation on more diverse microscopic images, typically levelling around 85%–89% accuracy. Moreover, although real-time object detectors such as YOLOv8 [13] can be deployed efficiently, they achieved only 83.00% overall accuracy on this dataset, with difficulty handling morphological overlap between ALL and HEM cells. These limitations are addressed by our Multi-model Ensemble Framework. Our model improves performance by combining deep semantic abstractions from ResNet50 with local texture descriptors from HOG and LDP, achieving 95.30% accuracy and a 96.50% F1-score. This performance is superior to the YOLOv8 baseline [13] and other recent hybrid approaches [8,9,19], demonstrating that a voting ensemble integrating handcrafted and deep features provides a strong balance of discriminative power and clinical reliability for automated leukaemia diagnosis.

5. Conclusion

This study presents a new ensemble-based deep learning framework for the exact identification of leukaemia from microscopic blood smear images. Apart from conventional architectures, including ResNet, and a Baseline CNN, the proposed method combines a Modified CNN enhanced with hybrid feature extraction techniques, such as histogram of oriented gradients (HOG), Local Directional Patterns (LDP), and deep features from ResNet50, into an ensemble model. Following extensive benchmark data analysis, the Proposed Ensemble Model performed better with 95.30% accuracy, 96.30% precision, 96.70% recall, and an F1 score of 96.50%. These values were somewhat better than those of individual models such as 85.47% accuracy, 83.81% precision, 97.08% recall, and 89.82% F1-score for the modified CNN, ResNet: 91.00% F1-score, 91.15% recall, 77.70% precision, and 88.50% accuracy. Baseline CNN: 75.00% precision, 77.00% accuracy, 84.00% recall, 79.5% F1-score. Training curve analysis covering 50 epochs further supports the durability and fast convergence of the proposed model. While the Baseline CNN models showed initial instability or slower convergence, the modified CNN and ResNet models displayed enhanced learning behaviour. This group found the most consistent and accurate findings. Fascinatingly. The model's hybrid feature fusion and ensemble method enabled its capacity to balance discriminative power with generalisation capability, allowing it to be successful in automated leukaemia identification.

CRediT authorship contribution statement

Krishna Kumar Jha: Writing: review & editing, Visualization, Validation, Supervision, Methodology, Investigation, Conceptualization. **Karan Singh:** Writing: original draft, Visualization, Validation, Software, Methodology, Data curation, Conceptualization. **Suparna Karmakar:** Writing: original draft, Validation, Methodology.

Ethical statement

This study does not involve human participants, human data, or animal subjects. The datasets used in this research are publicly available and were used strictly for research purposes. Therefore, ethical approval and informed consent were not required for this study.

Funding

This research received no external financial or non-financial support.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors acknowledge the use of ChatGPT (OpenAI) as a language support tool for checking the grammar and improving the readability of the manuscript. The authors affirm that the intellectual content, study design, data analysis, and conclusions of this work are their own. We greatly appreciate the input, insights, and feedback of our mentor from this domain, which greatly shaped the direction of this work.

Data availability

The dataset generated and analysed in the current study is available from the authors.

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